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Ecophysiology of the Antarctic sea star *Odontaster validus*: Study of isotopic fractionation and tissue turnover under controlled conditions

Anne Vanhove

Under the supervision of Dr Loïc Michel

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1) Introduction

Odontaster validus is a sea star belonging to the family Odontasteridae. It is the most abundant sea star in Antarctica, found exclusively around this continent and in the Southern Ocean, at depths of up to 900 meters (Zenteno-Devaud et al., 2022). It is an omnivorous organism that feeds on decaying matter, algae, crustaceans, and occasionally other echinoderms (Kidawa, 2001). To date, limited knowledge is available regarding the ecophysiology of this species due to the remote habitat in which it lives.

As part of a project conducted in 2018, sea stars were reared in captivity for nine months under a controlled diet. They were then euthanized and dried in an oven at 60°C. This was carried out to measure two key parameters for the study of isotopic fractionation: the isotopic fractionation between the animal and its food source, and the isotopic tissue turnover rate. Isotopic fractionation refers to the difference in isotopic composition between an organism and what it consumes. This fractionation results from biochemical processes that occur, among other things, during food digestion and nutrient assimilation (Fry, 2006). The isotopic tissue turnover rate refers to the rate at which tissues acquire the isotopic signature of the organism's diet. Measuring turnover makes it possible to determine how long the diet influences the isotopic composition of an organism's tissues (Carter et al., 2009; Vander Zanden et al., 2015). In trophic ecology, these two parameters are essential for reconstructing the diet of an organism and the food webs to which it belongs.

The most commonly used stable isotopes in biology are those of carbon (^{13}C) and nitrogen (^{15}N), although oxygen (^{18}O), hydrogen (^{2}H), and sulfur (^{34}S) isotopes can also be used.

Carbon isotopes help identify the primary source of energy in an organism's food chain. They make it possible to determine whether a trophic network is based on terrestrial or aquatic primary producers, such as phytoplankton or macroalgae.

Nitrogen isotopes are used to determine the trophic position of an organism. The $^{15}N/^{14}N$ isotopic ratio increases with trophic level. Thus, a grazer has a higher $\delta^{15}N$ value than a phytoplankton-based organism but a lower value than a carnivore.

These measurements make it possible not only to determine the trophic position of an individual but also its dietary plasticity. This refers to the ability of an organism to shift its diet or food sources depending on resource availability.

Finally, oxygen, hydrogen, and sulfur isotopes can also be used depending on the objectives of the study. Sulfur provides complementary information to carbon and nitrogen by distinguishing between different dietary sulfur sources. Meanwhile, $\delta^{18}O$ and $\delta^{2}H$ provide insights into the geographic origins of organisms, for example in the context of migration studies.

In the case of *O. validus*, these isotopic measurements will allow the study of isotopic fractionation between the sea stars and their food. This is only possible because the specimens were reared and fed under controlled laboratory conditions, using food sources with known isotopic compositions. The results of this study will make it possible, in the future, to accurately determine the food sources of organisms sampled in the wild and to investigate other aspects of their diet, such as the presence of dietary plasticity.

2) Description of the host institution

2.1) University of Liège

The University of Liège was founded in 1817 by William I of the Netherlands. It is a public, French-speaking university located in the province of Liège. Each year, it welcomes nearly 28,000 students, supervised by more than 630 professors and 4,000 researchers. A wide range of programs is offered across various fields such as arts, law, medicine, and sciences. For over 70 years, the University of Liège has gained international recognition in the fields of biology and oceanography, notably thanks to the presence of the Liège Aquarium and the STARESO research station located in Corsica.Numerous research activities are conducted within the university in oceanography, marine biology, and the impacts of climate change on marine ecosystems. The FOCUS research unit at the University of Liège is specifically dedicated to these areas of study.

2.2) FOCUS Research Unit

The research unit is called FOCUS, which stands for “Freshwater and OCeanic science Unit of reSearch.” It focuses on the comprehensive study of aquatic ecosystems, including their biological, physicochemical, and biogeochemical components. Various approaches are used, such as ecosystem modeling, the study of anthropogenic impacts on aquatic environments, and the analysis of interactions between organisms and their environment. The researchers within this unit are specialized in a wide range of disciplines, including modeling, ecology, behavioral sciences, chemistry, and ecotoxicology. My supervisor, Mr. Loïc Michel, leads the Laboratory of Systematics and Animal Diversity. His research covers several areas, including the study of trophic networks and their dynamics, animal feeding ecology, and ecological plasticity. His work mainly focuses on marine invertebrates and their role in the functioning of polar and deep-sea ecosystems. To achieve this, he uses approaches based on trophic markers such as stable isotopes. Stable isotope analysis allows the reconstruction of an organism's feeding habits as well as its position within the trophic network. This is made possible by predictable variations in isotopic ratios along the food web, enabling the identification of both the trophic level and the diet of an organism.

3) Internship objectives

At present, the isotopic fractionation between *O. validus* and its food source, as well as the isotopic tissue turnover rate, remain unknown. This is largely due to the fact that its metabolism is still poorly studied and therefore not well understood. However, it is suggested that it is strongly influenced by environmental drivers, particularly seasonal dynamics and low water temperatures.

Indeed, isotopic turnover is primarily controlled by metabolism and growth rate, both of which are influenced by environmental temperature. As a result, organisms living in cold environments such as Antarctica are expected to have slower metabolisms and therefore longer isotopic turnover times (Ziegler et al., 2023). However, Antarctica is characterized by strong seasonal variability, particularly in terms of food availability (Brockington & Clarke, 2001). It has been shown that organisms living in such extreme environments may experience variations in their growth rates and their metabolism may vary throughout the seasons, becoming faster during periods of high productivity (Brockington et al., 2001; Sporta Caputi et al., 2024). These changes in metabolism and growth rate also affect tissue turnover, which becomes faster during these periods. Thus, some organisms living in extreme environments, such as Antarctica in the case of this study, have the capacity for a certain degree of metabolic plasticity, allowing them to respond rapidly to sudden environmental changes (Sporta Caputi et al., 2024).

In this context, during my internship, I was responsible for preparing *O. validus* samples (grinding, acidification, weighing) for isotopic analysis. I was also introduced to isotopic instrumentation, particularly the elemental analyzer coupled with isotope ratio mass spectrometry (EA-IRMS).

4) Detailed report of the activities carried out

4.1) Sample grinding

During my first week of the internship, I ground the samples using a ceramic mortar and pestle (**Figure 1**). First, the food samples were ground. These samples were labeled as follows: ALGAE_1_DEC_18 and ALGAE_2_DEC_18 for the algae, and FISH_1_ODO_FOOD_DEC_18, FISH_2_ODO_FOOD_DEC_18, and FISH_3_ODO_FOOD_DEC_18 for the fish muscle samples. Next, the sea stars were ground and divided into two groups. The first group consisted of samples numbered from 1 to 30, and the second group included samples labeled from T4_01 to T4_255. Each ground sample was placed in a glass container and labeled accordingly. The instruments were cleaned with water and acetone between each sample to prevent any cross-contamination.

Figure 1: Steps of sea star sample grinding. 1) Sample selection; 2) Grinding; 3) Transfer into containers and labeling

4.2) Acidification

To acidify the samples, the glass containers were placed in a chamber containing beakers filled with 37% hydrochloric acid. At this concentration, hydrochloric acid becomes volatile, allowing the samples to be acidified through vapor, which prevents their degradation (Lorrain et al., 2003). The samples remained in hermetically sealed chambers for 48 hours. Handling hydrochloric acid requires the use of gloves and safety goggles, and all procedures must be carried out under a fume hood.

It is important to monitor this step carefully, as some samples may require a longer exposure time depending on the amount of material in the glass containers. To verify that $CaCO_3$ has been fully dissolved, a small amount of ground sample can be placed in hydrochloric acid. If acidification is incomplete, contact with the acid will release carbon dioxide in the form of bubbles. In such cases, the samples must be returned to the chamber for an additional 24 hours. Once acidification is complete, the samples are placed in an oven at 60°C for 24 hours.

Acidification is a crucial step because sea stars possess an internal skeleton composed of calcium carbonate ($CaCO_3$). This skeleton contains inorganic carbon, which has a different isotopic origin from organic carbon. By acidifying the samples, this inorganic carbon is removed, leaving only organic carbon, thus ensuring accurate isotopic carbon analysis. Indeed, only organic carbon reflects the dietary inputs of the studied organism, whereas inorganic carbon originates from abiotic sources and does not provide information about trophic position or feeding behavior.

<u>4.3) Weighing into capsules</u>

The capsules used are made of tin, a material that provides flexibility, allowing them to be easily sealed, while also acting as a catalyst to ensure optimal combustion. An equal amount of tungsten and either sample or standard was added to each capsule. The standards used included sulfanilic acid, IAEA-N1, IAEA-C6, IAEA-SO5, and a seabass replicate. Sulfanilic acid is used as a reference material for sulfur, nitrogen, and carbon, allowing instrument calibration. IAEA-N1 consists of ammonium nitrate and has a $\delta^{15}N$ value of 0.4‰. IAEA-C6 is composed of sucrose with a $\delta^{13}C$ value of −10.8‰. IAEA-SO5 consists of barium sulfate with a $\delta^{34}S$ value of +0.5‰. IAEA-N1, C6, and SO5 are international standards used in isotope ratio mass spectrometry, enabling calibration and validation of nitrogen, carbon, and sulfur isotopic analyses, respectively. The seabass replicate, although not a certified standard, has a biochemical composition very similar to that of the studied samples and is therefore used as a secondary standard. Tungsten acts as a catalyst, ensuring optimal combustion of all materials.

Regarding quantities, blanks contained approximately 2.5 mg of tungsten. Sulfanilic acid ranged between 0.35 and 0.45 mg, IAEA-SO5 between 0.3 and 0.4 mg, while IAEA-C6 and IAEA-N1 were both around 0.5 mg. The seabass replicates were approximately 2.5 mg per capsule.

At the beginning of each batch, two blank capsules were placed, followed by five capsules of sulfanilic acid, then one capsule of IAEA-N1, one of IAEA-C6, one containing IAEA-SO5, and one containing the seabass replicate. Subsequently, approximately 2.5 mg of each *O. validus* sample was placed into capsules. Every 12 capsules containing *O. validus* samples were followed by one capsule with the seabass replicate, one with sulfanilic acid, and one containing IAEA-SO5. This arrangement allows verification of measurement accuracy and proper instrument calibration during isotopic analysis. In total, four series of 12 samples were prepared for each batch. After the fourth series, the sequence included one capsule of seabass replicate, one of IAEA-N1, one of IAEA-C6, one of IAEA-SO5, followed by two sulfanilic acid capsules and two blank capsules.

In total, four batches were prepared using 96-well plates. The first three batches each contained 48 *O. validus* samples, while the last batch included 2 algae samples, 3 fish muscle samples, and 16 *O. validus* samples. Overall, 165 out of 291 samples were placed into capsules for isotopic analysis. The remaining samples were ground and acidified but not encapsulated due to a general strike in delivery services, which delayed the arrival of the glass containers.

An Excel file was created with one sheet per batch, each containing two columns: one for the well composition and the other for the mass of material placed in each capsule.

4.4) Stable isotope measurements

The samples will be subjected to isotopic ratio measurements after my departure. These measurements are carried out using an EA-IRMS system, which consists of an elemental analyzer (EA) and an isotope ratio mass spectrometer (IRMS). First, the capsule undergoes combustion or pyrolysis to convert the solid sample into simple gases. These gases then pass through a reduction column, which allows only the gases of interest to be retained. They subsequently undergo gas chromatography, which separates the different gases before they are transported to the mass spectrometer by a helium flow, where isotopic ratios are measured.

The stable isotopes used in this study are carbon (^{13}C), nitrogen (^{15}N), and sulfur (^{34}S). Carbon isotopes will help identify the primary energy source in the food chain of *O. validus*, making it possible to determine whether its trophic network is based on terrestrial or aquatic primary producers, such as phytoplankton or macroalgae. Nitrogen isotopes will allow the determination of the trophic position of *O. validus* (predator, grazer, etc.). As for sulfur isotopes, they will provide additional insight into the origin of its diet. For instance, a diet mainly based on organic matter from sediments would result in lower $\delta^{34}S$ values compared to a more pelagic-based food web (e.g., phytoplankton).

5) Summary of the experience gained (scientific, social, professional, etc.)

From a professional perspective, this internship provided valuable insight into laboratory work and its challenges. Throughout the internship, I was involved in several technical steps that allowed me to develop practical skills while discovering the demands of laboratory work. Some tasks, such as grinding sea star samples and weighing into capsules, proved to be repetitive and time-consuming. However, these are essential steps for isotopic analysis. To make these tasks more engaging, I learned to alternate between different activities, such as grinding, capsule preparation, literature review, and Excel file creation. This organization helped maintain my motivation and made my work more dynamic.

Weighing samples into capsules requires precision and concentration, as it is a key step that determines the quality of future results. Although this type of meticulous work may not appeal to me in the long term, it allowed me to develop technical skills and rigor that are essential in scientific work. Additionally, it can be enjoyable to occasionally engage in more focused and calm tasks.

I also became familiar with techniques used in isotopic analysis. For example, I learned how to check and set up a reduction column for the EA-IRMS system with the help of Mr. Lepoint. Due to the short duration of the internship, I was not able to directly participate in all stages of the process. However, I had the opportunity to analyze isotopic data as part of a course taught by Mr. Gilles Lepoint and Mr. Loïc Michel ("Study of stable isotopes in marine environments"). I particularly enjoyed working with data in R and interpreting the results. This internship reinforced my interest in both sample collection and data analysis.

In parallel, teamwork management was an enriching experience. Although the internship was initially individual, coordination with other students was necessary, particularly due to the limited number of balances available. This organization allowed me to develop and strengthen my skills in communication and time management. It highlighted the importance of cooperation and mutual support in optimizing work within a shared environment.

In conclusion, this internship allowed me to explore new aspects of scientific work, from sample preparation to data analysis. I developed key skills such as communication, rigor, and precision, which are essential in the field of science.

6) Self-assessment of the internship (experience and career perspective)

During my first Master's degree, I completed a nine-month thesis internship in the marine biology laboratory at the Catholic University of Louvain-la-Neuve. Therefore, I already had an idea of how such an internship would proceed. I was not surprised by delays in material delivery or by the limited availability of equipment. This experience confirmed that such logistical challenges are common across laboratories.

Regarding the schedule, no fixed working hours were imposed, which provided flexibility in managing my time. This allowed me to develop time management skills and to find a balance between personal and professional life.

During this internship, my main task was to prepare samples for future isotopic analysis through grinding, acidification, and capsule preparation. The first week was quite monotonous and not very instructive, as it was entirely dedicated to grinding 291 samples. In the following weeks, I was able to diversify my tasks, making my days more dynamic and stimulating. However, the duration of the internship was too short to complete the full isotopic data analysis process, which I would have liked to experience.

Despite its short duration, this internship was highly enriching both professionally and personally. I received a warm welcome, and both the laboratory technician and Mr. Michel and Mr. Lepoint were always available to answer my questions. Additionally, the presence of students from different Master's programs provided an opportunity to exchange ideas with future scientists from diverse backgrounds. This experience confirmed that the working environment is just as important as the research topic for professional fulfillment.

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